

Research Article

Analyses of the NAC Transcription Factor Gene Family in *Gossypium raimondii* Ulbr.: Chromosomal Location, Structure, Phylogeny, and Expression Patterns[□]

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Abstract

NAC domain proteins are plant-specific transcription factors known to play diverse roles in various plant developmental processes. In the present study, we performed the first comprehensive study of the NAC gene family in *Gossypium raimondii* Ulbr., incorporating phylogenetic, chromosomal location, gene structure, conserved motif, and expression profiling analyses. We identified 145 NAC transcription factor (NAC-TF) genes that were phylogenetically clustered into 18 distinct subfamilies. Of these, 127 NAC-TF genes were distributed across the 13 chromosomes, 80 (55%) were preferentially retained duplicates located in both duplicated regions and six were located in triplicated chromosomal regions. The majority of NAC-TF genes showed temporal-, spatial-, and tissue-specific expression patterns based on transcriptomic and qRT-PCR analyses. However, the expression patterns of several duplicate genes were partially redundant, suggesting the occurrence of sub-functionalization during their evolution. Based on their genomic organization, we concluded that genomic duplications contributed significantly to the expansion of the NAC-TF gene family in *G. raimondii*. Comprehensive analysis of their expression profiles could provide novel insights into the functional divergence among members of the NAC gene family in *G. raimondii*.

Keywords: Cotton; expression patterns; NAC; phylogeny; structure.

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Introduction

Transcription factors (TFs) are a class of proteins that regulate gene expression by binding to specific *cis*-acting elements in the promoters of target genes, and then activating or repressing their transcription (Riechmann et al. 2000; Wray et al. 2003). Transcriptional regulation plays an important role in controlling many critical biological processes during the development of plants, such as cellular morphogenesis, signal transduction, and responses to environmental stresses and pathogen attacks (Riano-Pachon et al. 2007; Paulino et al. 2009; Zhang

et al. 2011). The fine-tuned interaction between a TF and its target gene contributes to the complex transcriptional regulatory network that governs the major cellular pathways in plants in response to external (biotic and abiotic stress signals) and internal (intrinsic developmental processes) stimuli. To reconstruct the transcriptional regulatory network in a particular plant such as cotton, we need to first identify the general and specific TFs involved (Ravcheev et al. 2011).

Genome sequences for a number of economically important plants including cotton are now available, and the sequencing of many more plant genomes is currently underway. This

information provides a solid foundation for identifying TFs and their candidate target genes. Based on bioinformatics analysis, more than 50 different families of TFs have been identified in plants (Paulino et al. 2009; Zhang et al. 2011). Several databases specific for plant TFs are publicly available online (Riano-Pachon et al. 2007; Paulino et al. 2009; Zhang et al. 2011). The latest version of the Plant TF Database (PlantTFDB 2.0, <http://planttfdb.cbi.pku.edu.cn>) contains 53,319 putative TFs predicted from 49 species. The *Arabidopsis* (*Arabidopsis thaliana* (L.) Heynh.) genome encodes at least 1,550 TFs, which account for about 6.2% of its annotated genes (AtTFDB, Arabidopsis Transcription Factor Database, <http://arabidopsis.med.ohio-state.edu/AtTFDB>). For the diploid cotton species *Gossypium raimondii* Ulbr., about 6.6% of its genome has been found to encode more than 2,700 TFs (Wang et al. 2012). In the tetraploid cotton *G. hirsutum* L., 1,116 TFs have been identified that are classified into 50 families based on expressed sequence tags (ESTs) (Zhang et al. 2011).

NAC (NAM/ATAF/CUC) transcription factors (NAC-TFs) form a unique class of transcription factors found in a wide range of land plants. The NAC-TF family is one of the largest TF families in plants, with more than 100 members in *Arabidopsis* (Ooka et al. 2003; Tuskan et al. 2006), about 140 genes in rice (*Oryza sativa* L.) (Ooka et al. 2003), 101 genes in soybean (*Glycine max* (L.) Merr., Pinheiro et al. 2009) and 163 genes in *Populus* (Hu et al. 2010). In the tetraploid cotton *G. hirsutum*, 65 NAC-TFs have been identified based on ESTs (Zhang et al. 2011). There are 70 gene models in the *Carica papaya* L. NAC-TF family and 79 gene models in the *Vitis vinifera* L. NAC-TF gene family (<http://plntfdb.bio.uni-potsdam.de/v3.0/>, Paulino et al. 2009).

The common characteristics of NAC-TF proteins are a conserved NAC domain comprising about 150 amino acids in the N-terminus and a highly variable transcriptional regulation region in the C-terminus (Aida et al. 1997; Kikuchi et al. 2000; Ooka et al. 2003). Since the first NAC-TF gene, named NAM (NO APICAL MERISTEM), was identified in petunia during shoot apical meristem (SAM) formation of embryogenesis (Souer et al. 1996), NAC-TF genes have been implicated in a number of important biological processes during plant development. For instance, CUC1, CUC2 and CUP-SHAPED COTYLEDON are known to be involved in SAM formation (Duval et al. 2002), NAP (NAC-LIKE, ACTIVATED BY AP3/PI) is involved in floral morphogenesis in *Arabidopsis* (Sablowski et al. 1998), AtNAC2 is involved in lateral root development (He et al. 2005), AtNAM is associated with the embryo and SAM (Duval et al. 2002), ATAF1/2 (a membrane-bound NAC-TF gene) and CUC2 are involved in regulation of cell division (Kim et al. 2006), and RD26 is involved in hormone signal pathways (Fujita et al. 2004; ANAC036, Kato et al. 2009) and deposition of nutrient compounds in wheat grains (Uauy et al. 2006; Waters et al. 2009).

However, recent studies have suggested that a number of NAC-TF genes are also important regulators of plant abiotic stress and defense responses, such as drought in *Arabidopsis* (RD26, Fujita et al. 2004), salt tolerance in rice (ONAC045, Zheng et al. 2009), mechanical wounding (Collinge and Boller 2001) and responses against different pathogens (Wang et al. 2009). Additionally, several lines of evidence have indicated that a significant portion of NAC-TF genes may play critical roles in the formation of secondary cell walls in vascular plants (Demura et al. 2002; Kubo et al. 2005; Du et al. 2010). In *Arabidopsis*, a subset of closely related NAC domain proteins including NST1 (NAC Secondary Wall Thickening Promoting Factor, NST), NST2 and NST3 act as the main transcriptional regulators of secondary cell wall biosynthesis (Mitsuda et al. 2005, 2007; Mitsuda and Ohme-Takagi 2008). Although a number of NAC-TFs have been functionally studied in *Arabidopsis* and rice, the functions of a large majority of NAC members still remain unknown. Especially in *G. hirsutum*, the typical model for fiber development and cellulose biosynthesis, there have been only a few reports on the functional analysis of NAC-TFs.

In present study, we performed a genome-wide identification of NAC-TF genes in the diploid cotton species *G. raimondii* and revealed the reason for its expanded NAC-TF gene family of 145 members. We carried out detailed studies of the phylogeny, genomic organization, gene structure and expression profiles of the NAC-TF family. The expression patterns of thirty NAC-TF genes were also investigated by quantitative real-time RT-PCR (RT-qPCR) analysis.

Results

Identification of the NAC-TF gene family in *G. raimondii*

NAC-TFs comprise a large protein family. Each NAC-TF contains a highly conserved N-terminal DNA-binding domain and a variable C-terminal domain (Ernst et al. 2004; Olsen et al. 2005). We applied the hmmsearch program in HMMER to search against the annotated protein sequences of the *G. raimondii* genome to predict TFs. The HMM profiles (PF02365) were downloaded from PlantTFDB V2.0 (<http://planttfdb.cbi.pku.edu.cn>). Initially, 156 putative NAC-TF genes were identified. Based on previous research reports and the characteristics of the DNA-binding domain (DBD) of NAC-TFs, all 156 putative NAC-TF genes were manually inspected. We checked the sequence alignment and found 11 apparently fragmented sequences, which mostly had incomplete NAC domains. After their removal, 145 NAC-TFs were identified. These *G. raimondii* NAC-TF genes were designated GRNAC001 to GRNAC145 (Table S1). The number of NAC-TF genes in the *G. raimondii* genome was approximately 1.45-fold higher than in *Arabidopsis*, 1.45-fold

higher than in *Theobroma cacao* L. and 1.74-fold higher than in *C. papaya* (Riano-Pachon et al. 2007; Argout et al. 2011; Zhang et al. 2011). The NAC-TF genes identified in *G. raimondii* encoded proteins ranging from 115 to 1,120 amino acid (aa) residues in length, with an average of 361 aa. Interestingly, by blastp (E-value $1\text{E}-15$, identity $>40\%$) searching, we found that at least two *G. raimondii* NAC-TF genes were orthologs of genes in *Arabidopsis* (Table S1).

Genomic locations and duplication of NAC genes on the *G. raimondii* chromosomes

All 145 members of the *G. raimondii* NAC-TF family were located in the genome assembly (Figure 1A, Table S1) of the released *G. raimondii* genome sequence (<http://cgp.genomics.org.cn/page/species/index.jsp>); 127 NAC-TF genes were mapped to 13 chromosomes, whereas 18 genes were distributed across 16 unmapped scaffolds. The distribution of the *G. raimondii* NAC genes across the chromosomes was nonrandom (Figure 1A). Chromosomes 1 and 7 contained the largest number of NAC genes, with 18 each. In contrast, only four were annotated on chromosomes 2 and 3.

In the *G. raimondii* genome, clustering of NAC-TF genes was detected in several chromosomes (Figure 1B); approximately

20% of the NAC-TF genes (20/145) were located within eight clusters. In some chromosomal regions, NAC-TF genes were arranged in tandem repeats (Figure 2A,B, Table S1). For example, five NAC-TF genes (GRNAC001, GRNAC002, GRNAC003, GRNAC004, and GRNAC005) were within a cluster localized on a 30-kb segment of chromosome 1. GRNAC002, GRNAC016, GRNAC017, and GRNAC018 were organized in another cluster within an 18-kb fragment on the same chromosome, whereas GRNAC113, GRNAC114, GRNAC115 and GRNAC116 were arranged in a cluster localized with a 26-kb segment on chromosome 11 (Figure 1B, Table S1). Among the 20 NAC-TF gene clusters, four contained 10 tandem genes, two possessed three tandem genes, and another two possessed two tandem genes.

Recent studies have shown that the *G. raimondii* genome has undergone at least two rounds of genome-wide duplication and 2,355 syntenic blocks have been identified as well as 39 triplicated regions (Wang et al. 2012). Some chromosomes are highly fragmented, probably as a result of multiple rounds of duplication, diploidization and chromosomal rearrangement in the genome (Wang et al. 2012). By mapping the *G. raimondii* NAC-TF genes to the duplicated blocks that were established by Wang et al. (2012), we analyzed the relationship between the NAC-TF genes and genome-wide duplications. The distribution

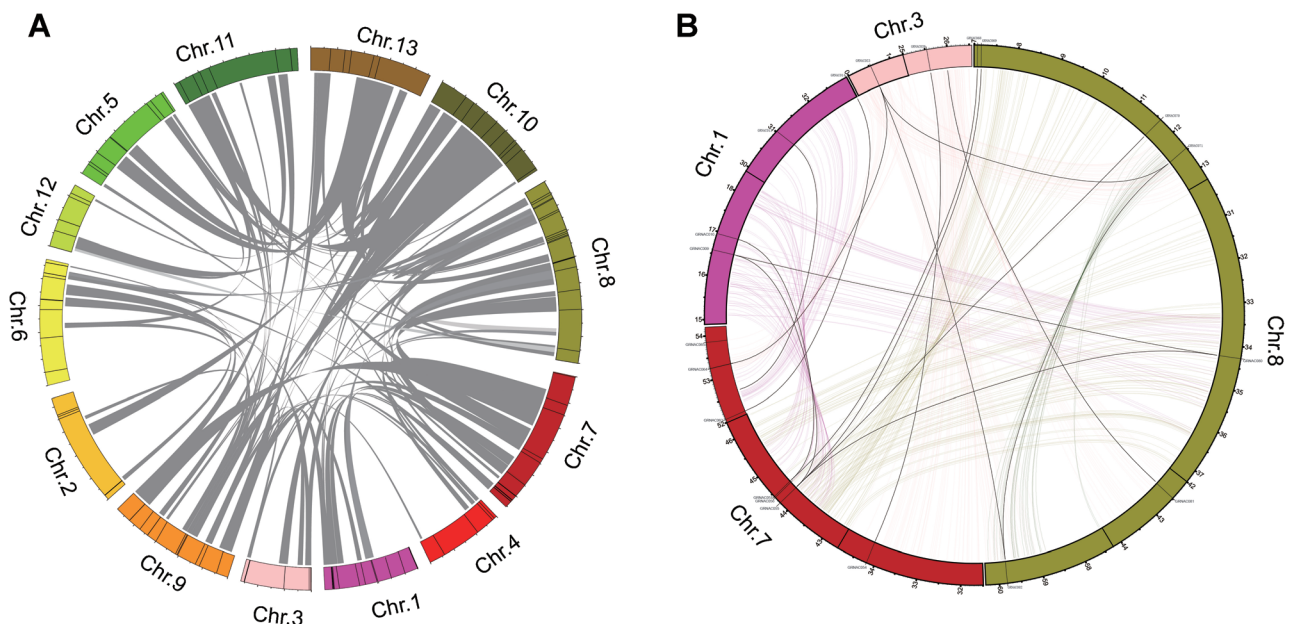


Figure 1. CIRCOS figure of reorganization of the *Gossypium raimondii* chromosomes.

(A) Syntenic blocks among different *G. raimondii* chromosomes. Only those syntenic blocks longer than 700 kb are shown.

(B) Detailed paralogous gene pairs of 12 duplicated blocks (Align2, Align3, Align36, Align37, Align38, Align39, Align40, Align67, Align72, Align73, Align74, and Align75) that were distributed on chromosomes 1, 3, 7, and 8.

Black lines represent paralogous pairs of *G. raimondii* NAC genes and other lines represent other paralogous gene pairs.

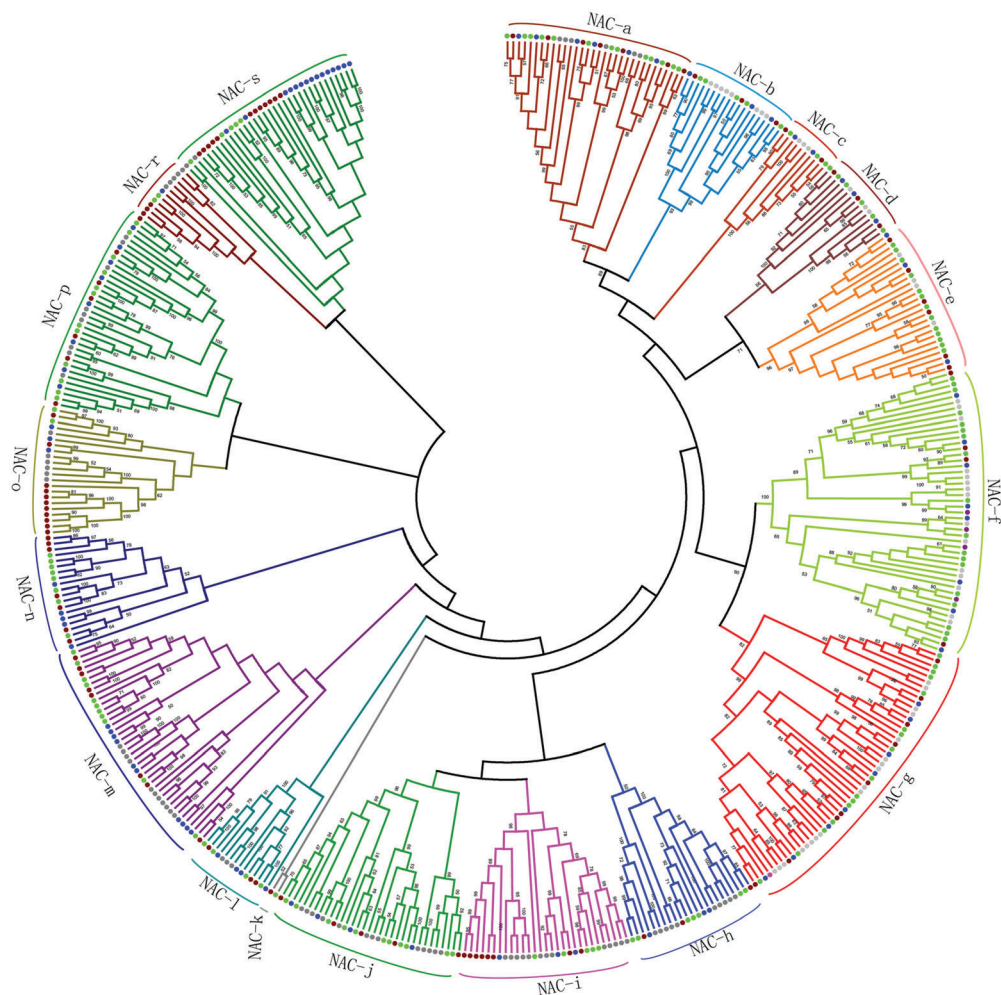


Figure 2. Phylogenetic tree of NAC domain-containing proteins from *Gossypium raimondii*, *Theobroma cacao*, *Arabidopsis* and rice.

The phylogenetic tree is based on a sequence alignment of the N-terminal NAM domains of 473 NAC protein sequences from four genomes, *G. raimondii*, *T. cacao*, *Arabidopsis*, and rice. The unrooted tree was generated with ClustalX using the neighbor-joining method. Bootstrap values from 5,000 replicates are indicated at each node. The NAC proteins are grouped into 18 distinct clades (a–s).

● *G. raimondii* ● *T. cacao* ● *Arabidopsis* ● Rice

of the NAC-TF genes in relation to the syntenic blocks is represented in **Figure 2A**. Ninety-two (76.37%) of the 127 chromosome-mapped *G. raimondii* NAC-TF genes were within 307 identified syntenic blocks, for example, GRNAC056 and GRNAC057 were located on chromosome 7, whereas GRNAC009 and GRNAC010 were located on chromosome 1 (**Figure 1B**). Thirty-three *G. raimondii* NAC-TF genes were located within 19 triplicated regions of the genome; these paralogous genes included GRNAC023, GRNAC082 and GRNAC071 (**Figure 2B**). In contrast, among the chromosome-mapped NAC-TF genes, 34 NAC-TF genes were located outside of any syntenic blocks (**Figure 1A**, Table S2).

Phylogenetic analysis of the NAC-TF gene family in *G. raimondii*, *T. cacao*, *Arabidopsis*, and rice

In the present study, a genome-wide approach was employed to predict NAC-TFs from different plant species. A total of 473 NAC-TF protein sequences from four genomes, *G. raimondii*, *T. cacao*, *Arabidopsis*, and rice, were predicted from alignments of the N-terminal protein domains of NAM. These NAC-TF proteins were classified into 18 subfamilies (namely NAC-a to NAC-s, **Figure 2**). A phylogenetic tree was constructed using MEGA 5.0 by employing the Neighbor-Joining (NJ) method. In a previous study, a similar phylogenetic analysis of NAC-TF family genes

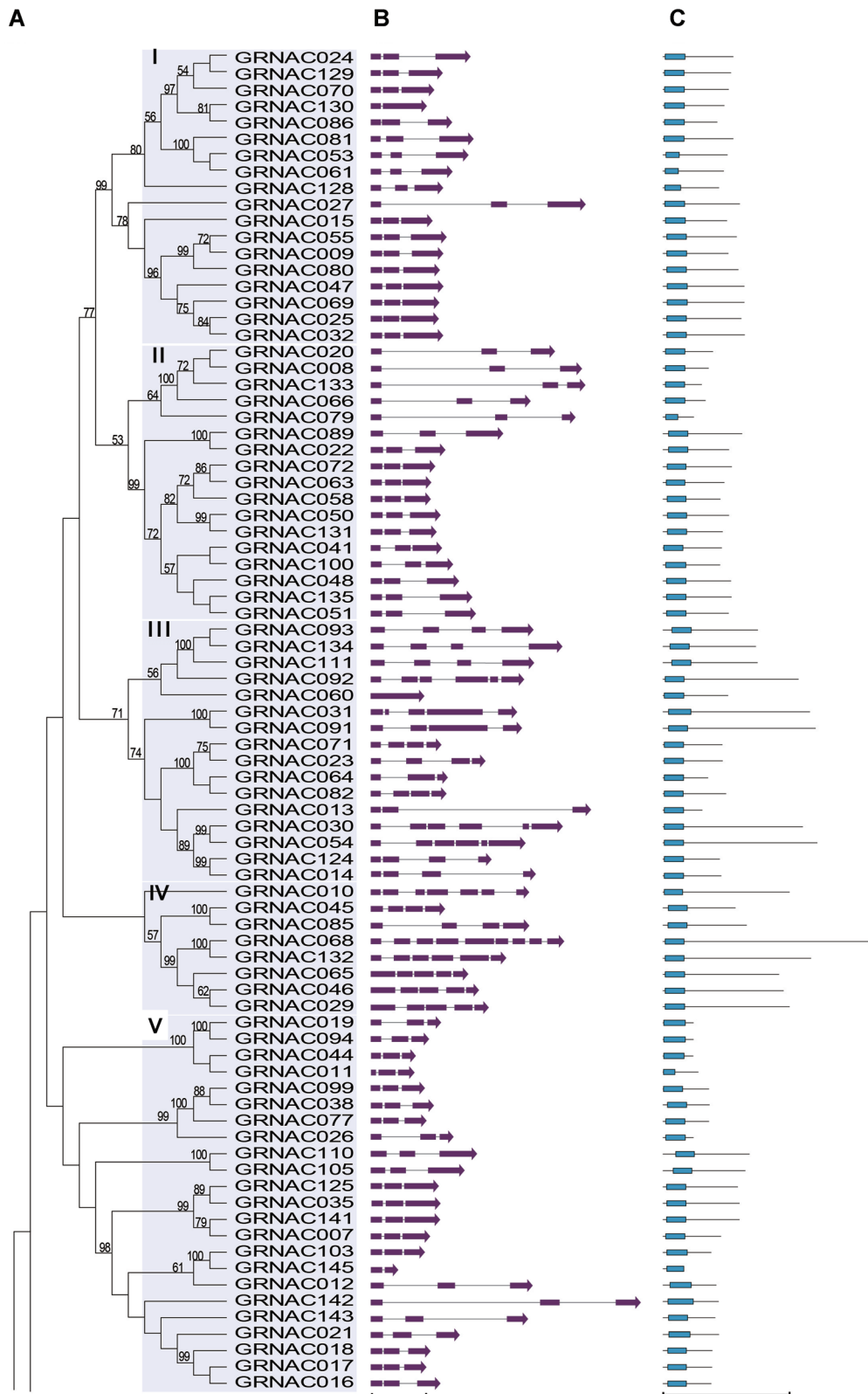


Figure 3. Continued.

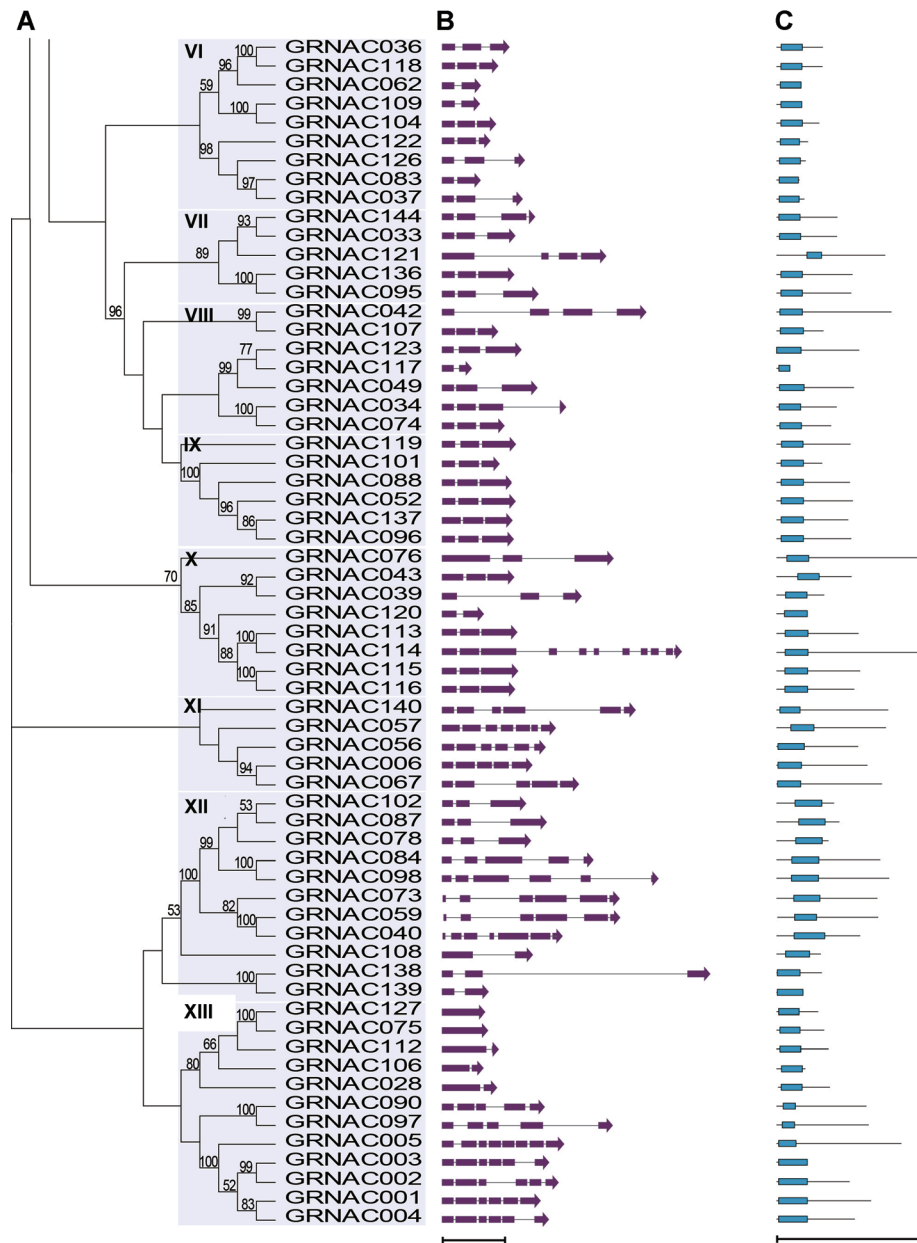


Figure 3. Phylogenetic relationships, gene structure and domain compositions of the *Gossypium raimondii* NAC genes.

(A) The phylogenetic tree was constructed with MEGA 5.0 using the Neighbor-Joining (NJ) method with 1,000 bootstrap replicates based on a multiple alignment of 145 amino acid sequences of NAC genes from *G. raimondii*. Bootstrap values higher than 50% are shown on the nodes. The thirteen major subfamilies are indicated with I to XIII and are marked with gray backgrounds.

(B) Exon/intron structures of NAC genes from *G. raimondii*. Exons and introns are represented by purple boxes and black lines, respectively.

(C) Schematic diagram of the conserved domain in NAC proteins from *G. raimondii* as annotated by Hmsearch. Black lines represent non-conserved sequences and blue boxes represent the NAM domain.

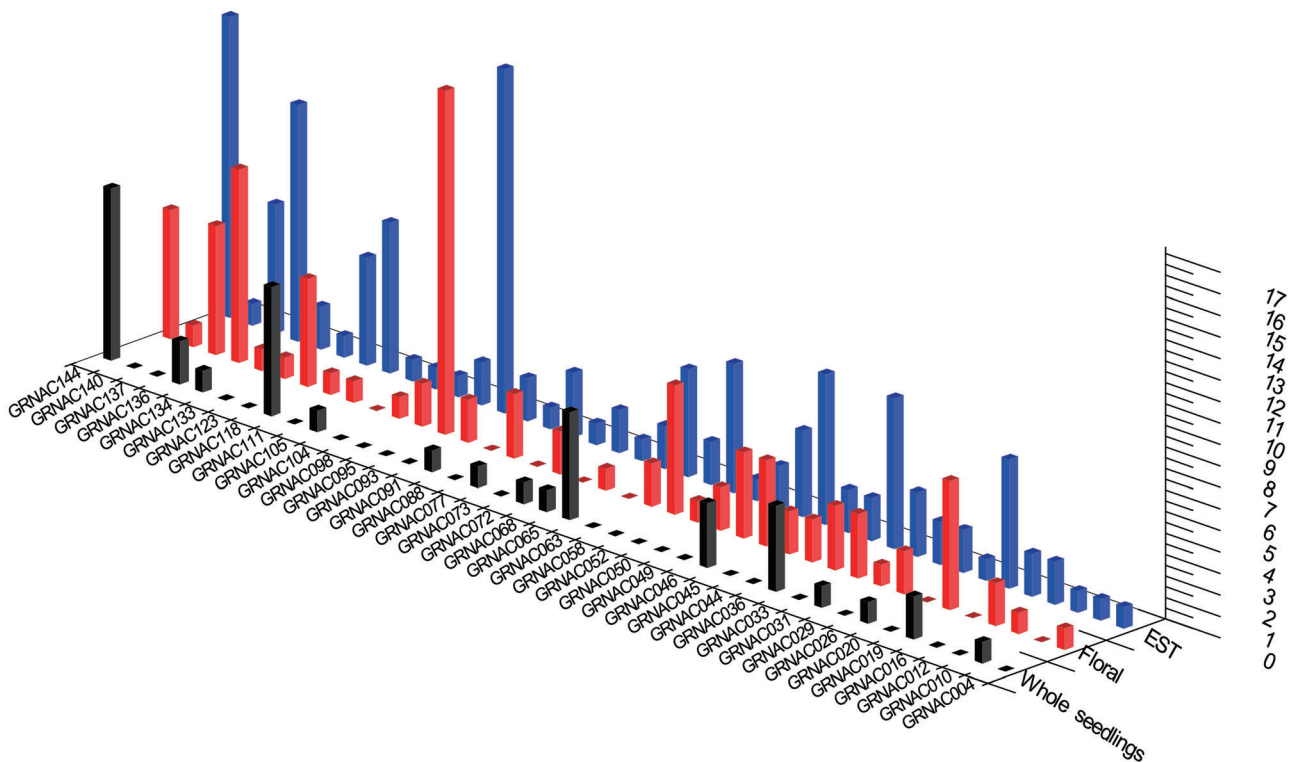


Figure 4. *In silico* frequency of *Gossypium raimondii* NAC genes ESTs.

The EST frequencies of 41 *G. raimondii* NAC genes were obtained by screening EST datasets from two libraries representing two tissues. The data for 68,732 *G. raimondii* ESTs were downloaded from the GenBank database.

from *Populus*, *Arabidopsis* and rice was performed with MEGA 5.0, and based on the phylogenetic tree, the NAC-TF family proteins were also divided into 18 distinct subgroups (Hu et al. 2010). GRNAC010 and Tc01_g033460 were distinguished from the other NAC-TF members and formed an individual clade with a bootstrap value of 99% (Figures 2, S1).

Gene structure and conserved domain analysis of *G. raimondii* NAC-TF genes

During the evolution of multigene families, diversification of gene structure is a common phenomenon, and may facilitate evolutionary co-option of genes for new functions to adapt to changes in the environment. By comparing the exon/intron structure in the coding sequences of the NAC-TF genes in *G. raimondii*, insights into the structural diversity of the NAC-TF genes were gained. The exon/intron structures are represented in Figure 3B and a NJ phylogenetic tree was generated for the 145 NAC-TF genes in *G. raimondii* that distributed the gene family into thirteen sub-clades (Figure 3A). Sub-clades I and II had 18 and 17 genes, respectively (Figure 3A). Subfamilies V,

VII, and IX had eighteen genes and the members in each subfamily possessed similar gene structure with two introns and similar exon lengths (Figure 3A, B). Comparatively, the gene structures of subfamilies X and XII varied greatly, with exon numbers ranging from 2 to 8 (Figure 3B).

The NAC subdomain homologs found in the HMM search are presented in a schematic diagram (Figure 3C). All of the NAC proteins in *G. raimondii* had the general structure of a NAC-TF protein, containing an N-terminal conserved NAC-domain (Figure 3C). Members belonging to the same subfamilies of the phylogenetic tree had similar gene structures and conserved motifs, which was consistent with the phylogenetic analysis.

Expression analysis of *G. raimondii* NAC genes

The *in silico* frequencies of *G. raimondii* NAC-TF genes and ESTs allows a basic statistical analysis of gene expression profiles in different tissues that have been deposited in public databases. Udall et al. (2006) published a global assembly of *Gossypium* ESTs. As part of this work, two large cDNA libraries from *G. raimondii* were released. These included 33,671 and

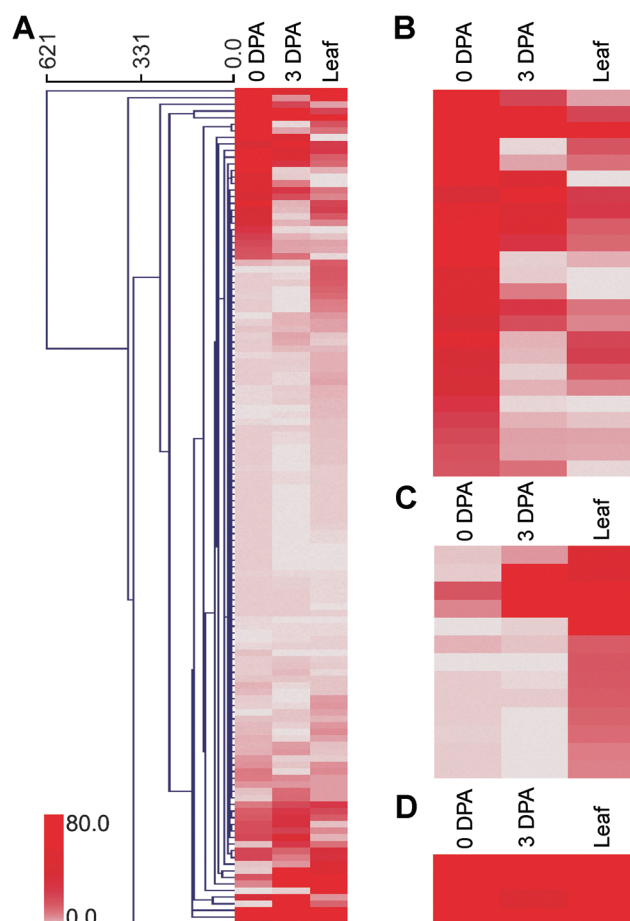


Figure 5. Expression patterns of the NAC gene family in *Gossypium raimondii*.

(A) Heatmap showing the clustering of 128 NAC-TF genes across three tissues (leaf, ovules at 0 DPA and 3 DPA; mentioned at the top of each lane). RNA seq data under the series accession number SRA048621 was obtained from the NCBI Sequence Read Archive (SRA) database.

(B) The expression levels of 28 NAC genes in 0- and 3-DPA ovules.

(C) Expression patterns of 13 NACs in dormant leaves.

(D) Expression profiles of 4 NACs in 0- and 3-DPA ovules and leaves.

The color scale at the bottom of the dendrogram shows the relative expression levels.

35,061 EST sequences obtained from the libraries GR_Ea (whole seedlings with first true leaves) and GR_Eb (3 DPA buds to +3 DPA bolls), respectively (Udall et al. 2005). We downloaded all 68,732 *Gossypium* ESTs from the GenBank database (<http://www.ncbi.nlm.nih.gov/est/>) to analyze the frequencies of the *G. raimondii* NAC-TF genes. The EST

frequencies are illustrated in **Figure 4** and Table S3. The expression profiles demonstrated that 41 NAC-TF genes were expressed in the two libraries; 32 genes were preferentially expressed in floral tissues and 11 NAC-TF genes were expressed in whole seedlings at the first true leaf stage.

To study the expression profiles of NAC-TFs genes in *G. raimondii*, we conducted transcriptome analysis using the MeV software (Saeed et al. 2003). Overall, 87.56% (127 of 145) of the identified NAC-TFs genes from the *G. raimondii* genome were supported by transcriptome sequencing data (RNA sequencing, RNA-seq; **Figure 5**, Table S4), and also by qualitative transcript analysis of data RNA-seq. Of these, 23 NAC-TFs were identified as up-regulated in ovules (0 DPA and 3 DPA) compared with leaves (**Figure 5B**, Table S4). The analysis also revealed that 13 *G. raimondii* NAC-TFs were up-regulated in leaves compared with ovules (0 DPA and 3 DPA). Interestingly, the expression of four NAC-TFs was detected in both ovules and leaves of *G. raimondii* (**Figure 5D**, Table S4).

Quantitative real-time RT-PCR analysis of expression profiles for 30 selected NAC-TF genes

To verify the expression profiles of *G. raimondii* NAC-TF genes obtained by EST frequency and transcriptome analysis, quantitative real-time RT-PCR (RT-qPCR) was performed on seven different tissues including ovules at -3, 0, 4, and 7 DPA, leaves, petals and bracts for 30 selected NAC genes. As illustrated in **Figure 6**, these genes showed distinct tissue-specific expression patterns. For example, 21 NAC-TF genes were strongly and preferentially expressed in petals. They were GRNAC009, GRNAC0021, GRNAC025, GRNAC026, GRNAC031, GRNAC034, GRNAC036, GRNAC037, GRNAC038, GRNAC046, GRNAC049, GRNAC057, GRNAC069, GRNAC080, GRNAC091, GRNAC093, GRNAC099, GRNAC123, GRNAC136, GRNAC142, and GRNAC144. A number of NAC-TF genes including GRNAC009, GRNAC021, GRNAC025, GRNAC033, GRNAC036, GRNAC050, GRNAC080, GRNAC136, and GRNAC142 were highly expressed in leaves, whereas other NAC-TF genes such as GRNAC033, GRNAC037, GRNAC068, GRNAC096, GRNAC134, GRNAC136, and GRNAC144 were highly expressed in bracts. These genes may play important roles in these tissues.

On the other hand, the expression of NAC-TF genes such as GRNAC021, GRNAC025, GRNAC031, GRNAC044, GRNAC046, GRNAC066, GRNAC072, GRNAC123, and GRNAC133 was down-regulated in 4 DPA ovules. The expression of NAC-TF genes such as GRNAC021, GRNAC025, GRNAC031, GRNAC044, GRNAC046, GRNAC057, GRNAC060, GRNAC091, GRNAC093, GRNAC099, and GRNAC0123 was down-regulated in 7 DPA ovules. The expression of a number of NAC-TF genes including GRNAC021,

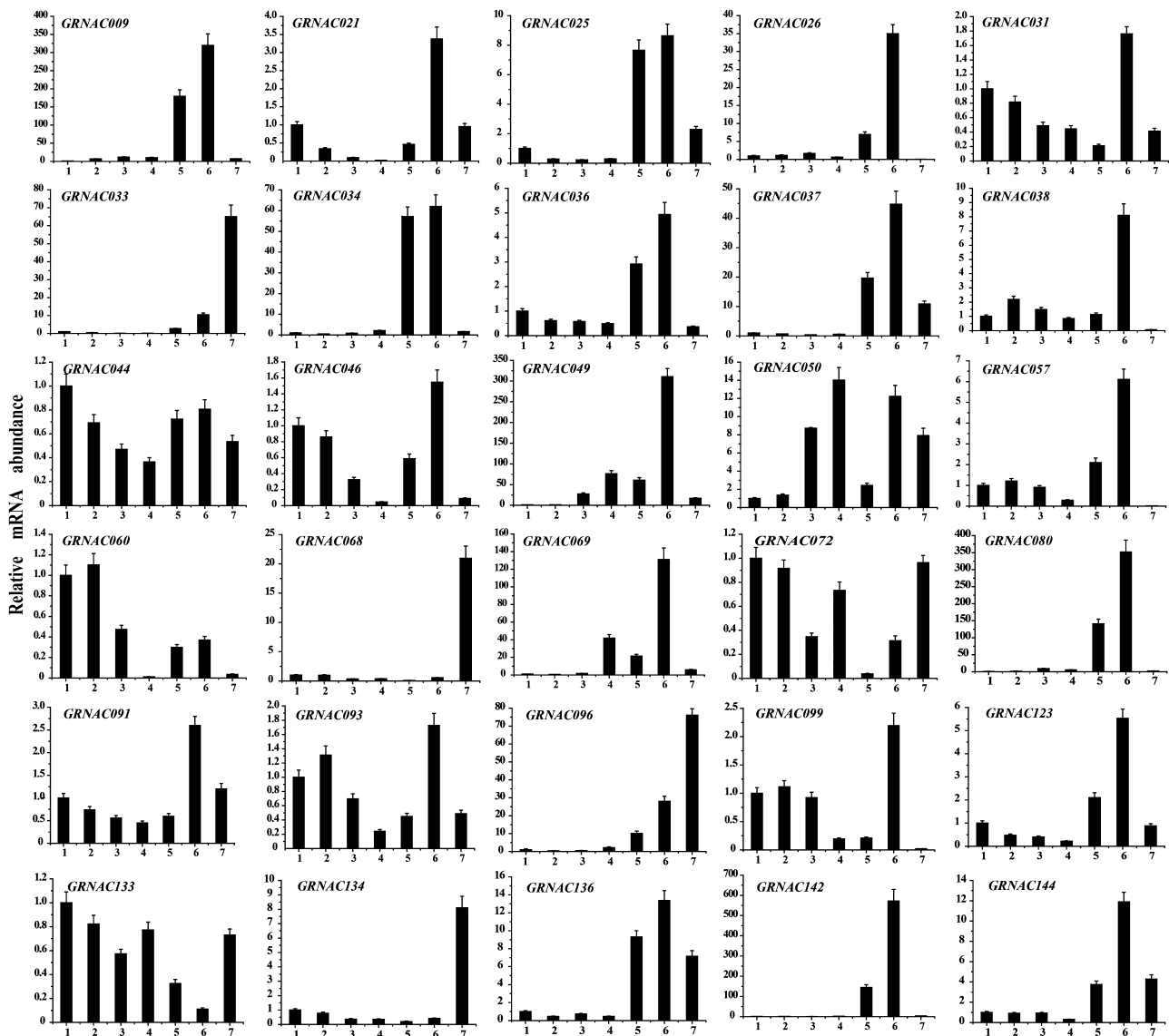


Figure 6. Expression analysis of thirty selected NAC genes using quantitative real-time RT-PCR (RT-qPCR).

The relative mRNA abundance of 30 selected NAC genes was normalized to the reference gene *histone 3* in different tissues. The bars show the standard deviation of three technical repeats. 1–4: –3, 0, 4, and 7 DPA ovules; 5: leaves; 6: petals; and 7: bracts.

GRNAC025, GRNAC031, GRNAC044, GRNAC046, GRNAC057, GRNAC060, GRNAC091, GRNAC093, GRNAC099, and GRNAC123 was down-regulated in leaves, while only the expression of GRNAC133 was down-regulated in petals. The expression of NAC-TF genes including GRNAC031, GRNAC046, GRNAC057, GRNAC060, and GRNAC099 was down-regulated in bracts.

Twelve NAC-TF genes were ubiquitously expressed among the different tissues; they included GRNAC031, GRNAC038, GRNAC044, GRNAC046, GRNAC050, GRNAC057, GRNAC060,

GRNAC072, GRNAC091, GRNAC093, GRNAC099, and GRNAC133. Several NAC-TF genes, including GRNAC033, GRNAC068, and GRNAC096, were preferentially expressed in subtending leaves. A subset of eight genes among the 30 NAC-TF genes tested, including GRNAC049, GRNAC050, and GRNAC069, were strongly expressed in different stages of ovules. Since only a few NAC-TF genes have been functionally characterized to date, the results (145 NAC-TF genes) presented here may provide some clues for the selection of candidate genes for further characterization.

Discussion

Expansion of the NAC TF gene family in *G. raimondii*

This study revealed that the NAC-TF gene family in *G. raimondii* is larger than in some other sequenced dicotyledonous angiosperms such as *Arabidopsis*, *T. cacao*, and *C. papaya*. The expansion of NAC-TF genes in the *G. raimondii* genome supports observations made by other investigators that many NAC-TFs play important roles in plant-specific processes. For instance, NAM, CUC1, CUC2, and AS2 have been shown to be involved in embryonic, floral and vegetative development (Souer et al. 1996; Aida et al. 1997; Duval et al. 2002). AIR3 and DBP, the downstream targets of NAC1, were reported to play a role in lateral root formation and auxin signal transduction (Xie et al. 2000). Other NAC-TFs were found to be involved in defense and abiotic stress responses (Paulino et al. 2009; Zhang et al. 2011) and in transcriptional regulation of secondary cell wall formation (Mitsuda et al. 2005, 2007; Mitsuda and Ohme-Takagi 2008). Previous studies revealed that *G. raimondii* has undergone the hexaploidization event shared by all eudicots and a cotton-specific whole-genome duplication (Wang et al. 2012) and that some chromosomes are highly fragmented, probably as a result of multiple rounds of duplication, diploidization and chromosomal rearrangement in the genome (Wang et al. 2012). By analyzing the relationship between sequence divergence within the NAC protein-encoding family and gene duplication in *G. raimondii*, we determined the chromosomal location of each NAC-TF gene in the *G. raimondii* genome (Figure 2A). For example, we found that GRNAC009, GRNAC055, and GRNAC080 were located in triplicated regions of chromosomes 1, 7, and 8, respectively. Our results (Figure 2, Table S1) suggest that the expansion of the NAC-TF gene family in *G. raimondii* arose from whole genome duplication events. The expansion of other transcription factor gene families in plants has been similarly associated with this process. For instance, the expansion of the F-box family in rice might have occurred in large part because of localized gene duplications (Jain et al. 2007). In *Populus*, duplicate NAC genes that are located in both duplicated regions have been preferentially retained (Hu et al. 2010).

In addition to the whole-genome duplication event in the *Gossypium* lineage, this expansion appears to have arisen from tandem duplication events (Figure 2A, Table S1). Interestingly, this finding is consistent with several previous reports in which a similar phenomenon was analyzed (Gagne et al. 2002; Nakano et al. 2006; Hu et al. 2010). For instance, the expansion of the *Populus* NAC-TF gene family appears to be the result of tandem duplication events (Hu et al. 2010). In *Arabidopsis* and rice, the expansion of the ERF family might have occurred through chromosomal/segmental duplication and tandem duplication (Nakano et al. 2006). In *Arabidopsis*, tandem duplication has

been implicated in the expansion of the F-box protein gene family (Gagne et al. 2002).

G. raimondii NAC-TF gene structure and conserved motifs

In the present study, we employed bioinformatic approaches to identify NAC-TF genes at a genome-wide level. A sequence phylogeny was constructed for 473 NAC-TF protein sequences from *G. raimondii*, *T. cacao*, *Arabidopsis*, and rice based on their N-terminal protein NAC domains. The NAC-TF proteins were classified into 18 subfamilies (NAC-a to NAC-s). The phylogenetic tree obtained for *G. raimondii*, *T. cacao*, *Arabidopsis* and rice paralleled those obtained in previous analyses (Ooka et al. 2003; Shen et al. 2010). For example, in phylogenetic analyses of NAC-TF proteins in *Arabidopsis* and rice (Ooka et al. 2003), and *Populus*, *Arabidopsis* and rice (Shen et al. 2010), the NAC-TF family proteins were divided into 18 distinct subgroups.

Functionalization of the duplicated genes

Duplicated genes can be maintained through sub-functionalization at the level of gene expression, which is indicated by divergence in their expression patterns (Duarte et al. 2006). In our research, a similar phenomenon was observed in RNA-seq analysis (Figure 5A). Twenty-three NAC-TFs showed increased transcript abundance during the development of ovules (0 DPA and 3 DPA; Figure 5B, Table S4) and 13 NAC-TFs exhibited increased transcript abundance in leaves (Figure 5C, Table S4). In addition, four NAC-TFs were expressed both in ovules and leaves (Figure 5D, Table S4). For example, the duplicated paralogous pairs (GRNAC010 and GRNAC056; GRNAC009, GRNAC055, and GRNAC080) showed different expression patterns (Figures 4, 5). The expression patterns of these duplicated paralogous pairs indicate that *G. raimondii* NAC genes have also been preserved by stable sub-functionalization. Furthermore, some *G. raimondii* NAC-TF genes, such as GRNAC009, GRNAC0026, GRNAC033, GRNAC068, GRNAC134, and GRNAC142, showed tissue-specific expression patterns, which were confirmed by qRT-PCR (Figures 5, 6). More interestingly, two homologous pairs (GRNAC009 and GRNAC080) also demonstrated almost identical patterns of transcript accumulation in leaves and petals (Figure 6). A similar phenomenon was observed in comprehensive analysis of the NAC-TF gene family in *Populus* (Hu et al. 2010).

NAC-TF proteins comprise one of the plant-specific families of transcription factors. The functional redundancy of NAC-TF proteins was first investigated by Vroemen et al. (2003), who reported that a third *Arabidopsis* NAC gene, CUC3, was involved in the establishment of the cotyledon boundary and the shoot

meristem. Our gene expression studies have shown that several members of the *G. raimondii* NAC-TF gene family, such as GRNAC044, GRNAC046, GRNAC050, and GRNAC133, have functional redundancy (Figures 4, 5). The redundancy in the *G. raimondii* NAC-TF gene family shows in the high similarity in transcript abundance profiles and is in line with the hypothesis that functional redundancy can be maintained for long periods, possibly because of an advantage conferred by genetic buffering (Zhang 2003).

Genetic analyses have revealed that a number of NAC protein transcription factors regulate secondary wall formation during fiber and vascular vessel development in *Arabidopsis* (Zhong et al. 2006; Demura and Fukuda 2007). These NAC TF proteins (NST1, NST2, and NST3/SND1) are clustered into one subfamily (Hu et al. 2010). Here, six functionally annotated *Arabidopsis* NST3/SND1 orthologs in *G. raimondii* (GRNAC032, GRNAC009, GRNAC025, GRNAC047, GRNAC055, and GRNAC069) fell into this subfamily, NAC-f (Figure 2, S1, Table S1). In previous reports, expression analysis showed that SND1/NST3 is expressed in the fiber cells of inflorescence stems and hypocotyls, and the valve endocarp layer and cells surrounding the vascular vessels in the replum of siliques in *Arabidopsis* (Zhong et al. 2006; Mitsuda et al. 2007), and that NST2 is expressed in the *Arabidopsis* anther endothecium (Mitsuda et al. 2005). In rice, OsSWN7 is expressed in the stem during cellular secondary wall formation (Zhong et al. 2011). In our research, GRNAC009, GRNAC069, and GRNAC025, which belong to the NAC-f clade, were highly expressed in leaves and petals in comparison with ovules. The similar expression patterns of these genes suggest that their products play important functional roles in *G. raimondii* development. Since only a few NAC-TF genes have been functionally characterized to date, our results (145 NAC-TF genes) may help in the selection of candidate genes for further characterization.

In summary, our comprehensive analysis of the NAC-TF gene family in *G. raimondii* led to the identification of 145 NAC-TF genes, which were phylogenetically clustered into 18 distinct subfamilies. Their gene structures and motif compositions were well conserved among the subfamilies. The majority of these genes were mapped to loci across the 13 chromosomes, with preferentially retained duplicates located in both duplicated regions and six members located in triplicated chromosomal regions. The majority of NAC-TF genes showed temporal-, spatial- and tissue-specific expression patterns. However, the expression patterns of some duplicate genes were partially redundant. These results suggest the occurrence of sub-functionalization during their evolution. Based on the genomic organization, we concluded that genomic duplications contributed significantly to the expansion of the NAC-TF gene family in *G. raimondii*. Comprehensive analysis of the expression profiles of all members of the NAC-TF gene family would provide novel

insights into the functional divergence among members of the NAC-TF gene family in *G. raimondii*.

Materials and Methods

Databases and sequences

We selected four plant species, *Arabidopsis* (*Arabidopsis thaliana* (L.) Heynh.), rice (*Oryza sativa* L.), *Theobroma cacao* L., and *Gossypium raimondii* Ulbr., for NAC-TF gene identification. The genome sequence of *G. raimondii* was downloaded from the Cotton Genome Project (<http://cgp.genomics.org.cn/>). The released *Arabidopsis* genome sequence was obtained from <http://www.arabidopsis.org/>. The Rice Annotation Project Database protein sequence was downloaded from <http://rapdb.dna.affrc.go.jp>. Sequencing data for the *G. raimondii* transcriptome was downloaded from the NCBI Sequence Read Archive (SRA, accession SRA048621).

RNA-seq analysis

Total RNA was isolated from all collected tissues (−3, 0, 4, and 7 DPA ovules, leaves, petals and bracts) using the Plant RNeasy extraction kit (Invitrogen, China). The integrity of total gDNA and RNA was determined by agarose gel electrophoresis, and the concentration and purity of the total RNA was assessed using a NanoDrop 2000c (Thermo Scientific, Wilmington, DE, USA). Normalized pools were converted to full-length enriched cDNA using the SMART method and cDNAs were sequenced using Illumina protocols. Clean reads were then mapped to the *G. raimondii* gene models using the CLC Genomics Workbench software 4 (CLC bio A/S Science) and matches were converted to RPKM to estimate gene expression levels. Expression data were analyzed and visualized with the MeV software (Saeed et al. 2003).

Quantitative RT-PCR

Based on the CDSs of the NAC-TFs, gene-specific primers for qRT-PCR were designed with the software Primer Premier 5.0 (Table S5). Total cDNA synthesis for the different cotton tissues (−3, 0, 4, and 7 DPA ovules, leaves, petals and bracts) was performed using the PrimeScript® RT Reagent Kit (Perfect Real Time). The expression of 30 NAC-TF genes was normalized using the mean value of the histone 3 gene (AF024716) as an internal control. All qRT-PCR reactions were run on an Applied Biosystems 7500 Fast Real-Time PCR System (Applied Biosystems) according to the manufacturer's protocol. Each 25 µL reaction volume included 12.5 µL of 2× Ultra SYBR Master Mix (with ROX), 0.5 µL of forward and reverse primers

(10 $\mu\text{mol L}^{-1}$), 2.5 μL of cDNA dilution, and 9 μL of ddH₂O. The thermal profile used for all PCRs was as follows: 10 min at 95°C for DNA polymerase activation, and 40 cycles of 15 s at 95°C and 35 s at 60°C. The expression levels were calculated using the $2^{-\Delta\Delta C_T}$ method. Each PCR was run in triplicate in each assay.

G. raimondii-specific NAC-TFs

NAC-TF family members in the *G. raimondii* genome were identified based on the NAC-TF domain structure. The HMM profiles (PF02365) were downloaded from PlantTFDB V2.0 (<http://planttfdb.cbi.pku.edu.cn>). NAC-TFs were internally annotated using the HMMER software for domain identification (an E-value of 0.01 was used as the cutoff). An E-value cutoff of ≤ 1.0 was applied to keep significant matches, which were then further analyzed.

EST Search

EST sequences of *G. raimondii* were downloaded from the GenBank EST database. The NAC protein sequences were searched against the ESTs using the E-value cutoff of $\leq 1\text{E}-2$; ESTs that were more than 98% identical to the query were retained for further analysis.

Analysis of Intron-Exon Structure

Gene structure information was parsed from the GFF file downloaded along with the genomic data and was used as the input for graphical display at the Gene Structure Display Server of Peking University, China (<http://gsds.cbi.pku.edu.cn/>).

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Supporting Information

Additional supporting information may be found in the online version of this article at the publisher's website.

Figure S1. Phylogenetic tree of NAC domain proteins from *G. raimondii*, *T. cacao*, *Arabidopsis* and rice.

Table S1. A list NAC gene family in *G. raimondii*.

Table S2. Genomic locations of NAC genes.

Table S3. EST frequencies of NAC genes.

Table S4. Expression of NAC genes family of *G. raimondii*.

Table S5. Primer sequences of the selected NAC genes for RT-qPCR analysis.